



[BIT703 Web Technologies](#) / [Contents](#) / [Topic 10: Server-side web application development with procedural and object-oriented database programming](#) / [Database operations in PHP](#)

Database and PHP 10m

With PHP, you can connect to and manipulate databases. Some of the databases that commonly used to connect with PHP are MySQL, PostgreSQL, Sybase, IBM Db2 and Oracle Database.

Among these, MySQL is the most popular database system, which is widely used with PHP. It is a database system that is available with server stacks such as XAMPP, Wampserver, etc. MySQL is used as a preferred database system because it is fast, reliable and easy to use. MySQL uses industry-standard SQL and runs on several platforms. In PHP, you can either use a standard procedural approach with functions or the OOP approach. For this section, we will demonstrate the procedural approach first, followed by the corresponding OOP version.

Connecting to MySQL using the procedural approach

To start, add the following code to a PHP file and save it as **config.php** in the document root (www or htdocs) of your web application. This configuration file contains four constants (values that do not change during the execution of a program).

As you have seen in an earlier topic, a constant is declared in PHP using the built-in define function. In the parentheses, you define the name of the constant (in uppercase letters) and its (fixed) value. The example below matches the default username and password of MySQL and the name of the restaurant database you created in a previous topic.



If you need a refresher about this, we recommend taking a look at the following pages:

[Skip to main content](#)

Connecting database to client side →

Because the database is running as part of the same deployment at the server, the host is localhost (you would need the URL here if the database were running on a different server).

```
<?php

// MySQL credential

define( "DBHOST", "localhost");

define( "DBUSER", "root");

define( "DBPASSWORD", "test");

define( "DBDATABASE", "db_rating");

?>
```

Once you have defined these constants in the config.php file, you can include them in any other PHP files that need to use them. To do this, you use the keyword 'include', which is shown below:

```
include "config.php";
```

For the examples in this topic, you will be writing database connection code inside PHP pages so you can test the code running on the server.

Before using a database, you need to:

1. Connect to it.
2. Check that the connection was 'successful'

Skip to main content 3. Use the connection to issue SQL, followed by closing the database connection.

The next code example uses the following PHP MySQLi functions to interact with the database:

- **mysqli_connect(string \$hostname, string \$username, string \$password, string \$database, int \$port, string \$socket)** – makes a connection to the selected database. The arguments have default values you can omit from the right-hand end, but the default database name is an empty string that you must replace.
- **mysqli_connect_errno()** – returns the error code from the last connection attempt.
- **mysqli_connect_error()** – returns the error message from the last connection attempt.
- **mysqli_get_host_info(mysqli \$mysqli)** – gets the host information from the connection (mysqli is a type of MySQL connection).
- **mysqli_close(mysqli \$mysqli)** – closes a previously opened database connection.

It is recommended to read through the PHP MySQLi functions resource so you are familiar with their operation and to be aware of other related functions. The documentation also includes many examples.

Note: Error numbers and messages are not from PHP but from the underlying database. If you are connecting to MySQL, see the following page for MySQL error numbers and messages: Client Error Message Reference (MySQL, n.d.).^{fnref:5fa6ca}

The version of MySQL might differ from your installation, but the errors are generally the same across versions.

Add the following to a PHP file and save it as dbconnectiontest.php in the document root of your web server. Then test it in the browser by using the URL <http://localhost/dbconnectiontest.php>.



[Skip to main content](#)



You will continue using this connection syntax and config.php for all the database activities.

Any code that interacts with the database must occur between connecting to the database and closing that connection.

[Skip to main content](#)

In the examples in this section, you will assume that a database is already connected. You will focus on the syntax for each database create, read, update and delete (CRUD) operation.



Thoughts on this page?

